```
import numpy as np
```

```
marks = np.array([70, 80, 65, 90, 75])
print(marks.dtype)
```

```
int64
```

```
print(marks.ndim)
```

```
1
```

```
print(marks.shape)
```

```
(5,)
```

```
print(marks.size)
```

```
5
```

```
marks
```

```
array([70, 80, 65, 90, 75])
```

```
marks[0]
```

```
np.int64(70)
```

```
print(marks[3])
```

```
90
```

```
n = len(marks)
for i in range(n):
  print(marks[i], end = " ")
```

```
70 80 65 90 75
```

```
print(marks[n-1])
```

```
75
```

```
print(marks[-1])
```

```
75
```

```
print(marks[-2])
```

```
90
```

```
marks[2] = marks[2] + 5
```

```
print(marks)
```

```
[70 80 70 90 75]
```

```
marks + 5
```

```
array([75, 85, 75, 95, 80])
```

```
marks = np.array([
    [70, 80, 90],
    [60, 75, 85],
    [88, 92, 95]
])

print(marks)
```

```
[[70 80 90]
 [60 75 85]
 [88 92 95]]
```

```
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```

```
np.int64(90)
```

```
print(marks[0][2])
```

```
90
```

```
marks[0,2]
```

```
np.int64(90)
```

```
# print all the digaonal elements of the 2d np array

marks
```

```
array([[70, 80, 90],
       [60, 75, 85],
       [88, 92, 95]])
```

```
row,col = marks.shape
```

```
A college stores marks of 4 students in 3 subjects: Maths, Science, and English.

import numpy as np

marks = np.array([
    [72, 85, 90],
    [65, 78, 82],
    [88, 91, 84],
    [70, 76, 80]
])

Perform the following:

Display the marks of the third student.
Display the Science marks of the second student.
Add 5 bonus marks to every student's English marks.
Change the Maths marks of the fourth student to 75.
Display the final marks array.
```

Double-click (or enter) to edit

```
marks = np.array([
    [72, 85, 90],
    [65, 78, 82],
    [88, 91, 84],
    [70, 76, 80]
])
```

```
# Display the marks of the third student.

marks[2]
```

```
array([88, 91, 84])
```

```
# Display the Science marks of the second student.

marks[1,1]
```

```
np.int64(78)
```

```
# Add 5 bonus marks to every student's English marks.
marks[:,2]
```

```
array([90, 82, 84, 80])
```

```
marks[:,2] = marks[:,2] + 5
```

```
marks
```

```
array([[72, 85, 95],
       [65, 78, 87],
       [88, 91, 89],
       [70, 76, 85]])
```

```
# Change the Maths marks of the fourth student to 75.
marks[3,0] = 75
```

```
print(marks)
```

```
[[72 85 95]
 [65 78 87]
 [88 91 89]
 [75 76 85]]
```

```
Display the final marks array.
print(marks)
```

```
  File "/tmp/ipykernel_1459/1013035091.py", line 1
    Display the final marks array.
            ^
SyntaxError: invalid syntax
```

Next steps: Explain error

```
A company records monthly sales for 3 branches.
The columns represent January, February, and March.

import numpy as np

sales = np.array([
    [120, 150, 180],
    [100, 130, 160],
    [200, 220, 250]
])

Perform the following:

Display the complete sales data of Branch 2.
Display the sales of all branches in February.
Increase the March sales of every branch by 10%.
The January sales of Branch 1 were entered incorrectly. Change them from 120 to 135.
Display the updated sales array.
```

```
sales = np.array([
    [120, 150, 180],
    [100, 130, 160],
    [200, 220, 250]
])
```

```
# Display the complete sales data of Branch 2.

print(sales[1])
```

```
[100 130 160]
```

```
# Display the sales of all branches in February.

print(sales[:,1])
```

```
[150 130 220]
```

```
# Increase the March sales of every branch by 10%.
# a + 10%of a
# a + 0.1a
# a(1+0.1) = 1.1*a
```

```
sales[:,2] = sales[:,2]*1.1
print(sales[:,2])
```

```
[198 176 275]
```

```
# The January sales of Branch 1 were entered incorrectly. Change them from 120 to 135.

sales[0,0] = 135
```

```
print(sales[0,0])
```

```
135
```

```
# Display the updated sales array.
print(sales)
```

```
[[135 150 198]
 [100 130 176]
 [200 220 275]]
```

An organization stores employee information as:

Salary | Experience

```
import numpy as np

employees = np.array([
    [30000, 2],
    [40000, 4],
    [50000, 6],
    [60000, 8]
])
```

Perform the following:

Display the information of the last employee using negative indexing.
Display the salary of the second employee.
Increase the salary of the third employee by 15%.
Add 1 year of experience to the first employee.
Display the final employee array.

```
employees = np.array([
    [30000, 2],
    [40000, 4],
    [50000, 6],
    [60000, 8]
])
```

```
# Display the information of the last employee using negative indexing.
employees[-1]
```

```
array([60000,     8])
```

```
# Display the salary of the second employee.
employees[1,0]
```

```
np.int64(40000)
```

```
# Increase the salary of the third employee by 15%.

# a + 0.15*a = a(1+0.15) = 1.15*a

employees[2,0] = employees[2,0] + 0.15*employees[2,0]
```

```
# employees[2,0] = employees[2,0]*1.15
# employees[2,0] *= 1.15
```

```
# Add 1 year of experience to the first employee.
employees[0,1] += 1
```

```
print(employees)
```

```
[[30000     2]
 [40000     4]
 [57500     6]
 [60000     8]]
```

A company has inventory information for 4 products across 3 warehouses.

Columns: Warehouse A, Warehouse B, Warehouse C

```
import numpy as np

inventory = np.array([
    [100, 120, 150],
    [80, 90, 110],
    [200, 180, 160],
    [50, 70, 90]
])
```

Perform the following:

Display the inventory of Product 3 across all warehouses.
Display the inventory of Warehouse B for all products.
Warehouse A receives 20 additional units of every product. Update the array.
The inventory of Product 4 in Warehouse C is incorrect. Change it to 100.
Display the final inventory.

```
inventory = np.array([
    [100, 120, 150],
    [80, 90, 110],
    [200, 180, 160],
    [50, 70, 90]
])
```

```
# Display the inventory of Product 3 across all warehouses.

print(inventory[2])
```

```
[200 180 160]
```

```
# Display the inventory of Warehouse B for all products.
print(inventory[:,1])
```

```
[120  90 180  70]
```

```
# The inventory of Product 4 in Warehouse C is incorrect. Change it to 100.

inventory[3,2] = 100
```

```
print(inventory)
```

```
[[100 120 150]
 [ 80  90 110]
 [200 180 160]
 [ 50  70 100]]
```

A weather application stores temperatures for 3 cities.

Columns: Morning, Afternoon, Evening

```
import numpy as np

temperature = np.array([
    [24, 35, 28],
    [22, 32, 26],
    [27, 38, 30]
])
```

Perform the following:

Display the complete temperature data of the first city.
Display all afternoon temperatures.
Display the evening temperature of the last city using negative indexing.

```
Increase all morning temperatures by 2°C.
The afternoon temperature of the second city was recorded incorrectly. Change it from 32 to 34.
Display the final temperature array.
```

```
temperature = np.array([
    [24, 35, 28],
    [22, 32, 26],
    [27, 38, 30]
])
```

```
# Display the complete temperature data of the first city.


temperature[0]
```

```
array([24, 35, 28])
```

```
# Display all afternoon temperatures.
temperature[:,1]
```

```
array([35, 32, 38])
```

```
# Display the evening temperature of the last city using negative indexing.

temperature[-1,2]
```

```
np.int64(30)
```

```
temperature[-1,-1]
```

```
np.int64(30)
```

```
# Increase all morning temperatures by 2°C.

temperature[:,0] += 2
```

```
# The afternoon temperature of the second city was recorded incorrectly. Change it from 32 to 34.
temperature[1,1] = 34
```

```
print(temperature)
```

```
[[26 35 28]
 [24 34 26]
 [29 38 30]]
```

```
Start coding or generate with AI.
```